

Intermediate Quantitative Methods

Introduction

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First Term 2026–27

European University Institute, SPS

- Kevin Munger and Sebastian Ramirez-Ruiz

- Kevin Munger and Sebastian Ramirez-Ruiz
- TA: Martín Alberdi

People and times

- Kevin Munger and Sebastian Ramirez-Ruiz
- TA: Martín Alberdi
- Seminar: Friday 15–17, Seminar Room 2

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- TA: Martín Alberdi
- Seminar: Friday 15–17, Seminar Room 2
- Lab: Monday 17–19, Seminar Room 2

Today

- Who I am

Today

- Who I am
- What this course is, and what it used to be

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- Starting grad school

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- What this course is, and what it used to be
- Starting grad school
- Where quantitative social science is

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- Where quantitative social science is
- Where it is going

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- Where quantitative social science is
- Where it is going
- Logistics

Who I am

- PhD in Politics, NYU (SMaPP lab)

Background

- PhD in Politics, NYU (SMaPP lab)
- Assistant Professor, Penn State

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- Assistant Professor, Penn State
- EUI: Chair in Computational Social Science

What I work on

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- Digital media and political behavior

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- Field and survey experiments

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- Metascience: how social science produces knowledge

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- Generations and youth gender polarization

This course

- Normal distribution and z-scores

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- Inference and hypothesis testing

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- Bivariate and multivariate regression

- Normal distribution and z-scores
- Inference and hypothesis testing
- Bivariate and multivariate regression
- Fixing regression problems

- Normal distribution and z-scores
- Inference and hypothesis testing
- Bivariate and multivariate regression
- Fixing regression problems
- Nested data, TSCS, binary outcomes

- Foundations: inference, and what a regression is for

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- Before estimation: estimands and measurement

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- Before estimation: estimands and measurement
- Model families: MLE, nested data, machine learning

- Foundations: inference, and what a regression is for
- Before estimation: estimands and measurement
- Model families: MLE, nested data, machine learning
- The frontier: new data, generative AI

Why the change

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- Ten weeks is short

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- Most mistakes happen before estimation
- Wrong question, wrong measure, wrong estimand

- Simulate a data-generating process

How we work

- Simulate a data-generating process
- Break an assumption

How we work

- Simulate a data-generating process
- Break an assumption
- Observe what happens

- Simulate a data-generating process
- Break an assumption
- Observe what happens
- R and Quarto – slightly intimidating, useful for reproducible workflows

Starting grad school

What changes now

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- Methods courses are a map, a sense of what's out there

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- “Methodology” socially constructed

- Read more than feels efficient

Practical advice

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- Talk to people outside your subfield

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- Grad school is FUN

- Competitive academic job market

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- Rewards for volume and novelty

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- Science needs null results and honest reporting

- Competitive academic job market
- Rewards for volume and novelty
- Science needs null results and honest reporting
- These pull in different directions

Where things are

The credibility revolution

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- Design over modeling

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- Experiments, natural experiments, DiD, RD

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- Experiments, natural experiments, DiD, RD
- “What is your identification strategy?”

The credibility revolution

- Design over modeling
- Experiments, natural experiments, DiD, RD
- “What is your identification strategy?”
- Causal claims as the high-status output

The replication crisis

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- Garden of forking paths (Gelman & Loken 2014)
- Responses: preregistration, registered reports, open data and code

Many analysts, one dataset

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- Breznau et al. (2022): 73 teams

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- Same data, same hypothesis

Many analysts, one dataset

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- Estimates ranged from negative to null to positive

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- Same data, same hypothesis
- Estimates ranged from negative to null to positive
- Researcher decisions as a source of uncertainty

What is a regression for?

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- Description

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- Description
- Prediction

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- Prediction
- Causal inference

What is a regression for?

- Description
- Prediction
- Causal inference
- Same model, different questions, different standards

Description is underrated

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- JQD:DM (Munger, Guess & Hargittai 2021)

Before estimation

- What exactly are we estimating? (Lundberg, Johnson & Stewart 2021)

Before estimation

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- Do our numbers measure our concepts?

Before estimation

- What exactly are we estimating? (Lundberg, Johnson & Stewart 2021)
- Do our numbers measure our concepts?
- Neither is fixed by a better estimator

Where things are going

- Digital trace data

New data

- Digital trace data
- Text, images, audio, video

New data

- Digital trace data
- Text, images, audio, video
- Platforms control access

- Digital trace data
- Text, images, audio, video
- Platforms control access
- Found data has its own errors (Lazer et al. 2014)

- The social world changes, especially online

Temporal validity

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- Findings about 2016 platforms may not hold today

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- Knowledge decays, sometimes quickly

- The social world changes, especially online
- Findings about 2016 platforms may not hold today
- Knowledge decays, sometimes quickly
- Replication over time as a research goal

Generative AI in the research pipeline

Generative AI in the research pipeline

- Writing and debugging code

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- Text annotation and classification

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- Literature search

Generative AI in the research pipeline

- Writing and debugging code
- Text annotation and classification
- Literature search
- Simulated respondents (Argyle et al. 2023, Bisbee et al. 2024)

- Outputs still need validation

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- Proprietary models change and disappear

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- LLM agents in online survey samples

- Outputs still need validation
- Proprietary models change and disappear
- LLM agents in online survey samples
- More papers is not more knowledge

What lasts

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- Asking precise questions

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- Measurement and validation

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- Simulation to understand methods

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- Asking precise questions
- Measurement and validation
- Simulation to understand methods
- Reproducible workflow

What lasts

- Asking precise questions
- Measurement and validation
- Simulation to understand methods
- Reproducible workflow
- Reading, and judgment

Logistics

- Block 1, weeks 1–2: inference and regression (KM)

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- Block 2, weeks 3–5: estimands, DAGs, measurement (SRR)

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- Block 3, weeks 6–8: MLE, nested data, machine learning

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- Block 2, weeks 3–5: estimands, DAGs, measurement (SRR)
- Block 3, weeks 6–8: MLE, nested data, machine learning
- Block 4, weeks 9–10: new data, generative AI

- Mondays, with Martín

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- R and the tidyverss

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- R and the tidyverss
- Short activities

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- Simulation from the first lab

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- R and the tidyverss
- Short activities
- Simulation from the first lab
- Assignments as Quarto documents that render start to finish

- Good at explaining errors and drafting code

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- You are responsible for any code you did not write

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- Test it on cases where you know the answer

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- You are responsible for any code you did not write
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- Week 10 takes this up properly

Requirements

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- Attendance (80%) and participation

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- Short simulation and analysis assignments

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- Short simulation and analysis assignments
- Final exam